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UNIVERSITÀ DEGLI STUDI DI MILANO

Graph Neural Networks for Ambient Intelligence

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

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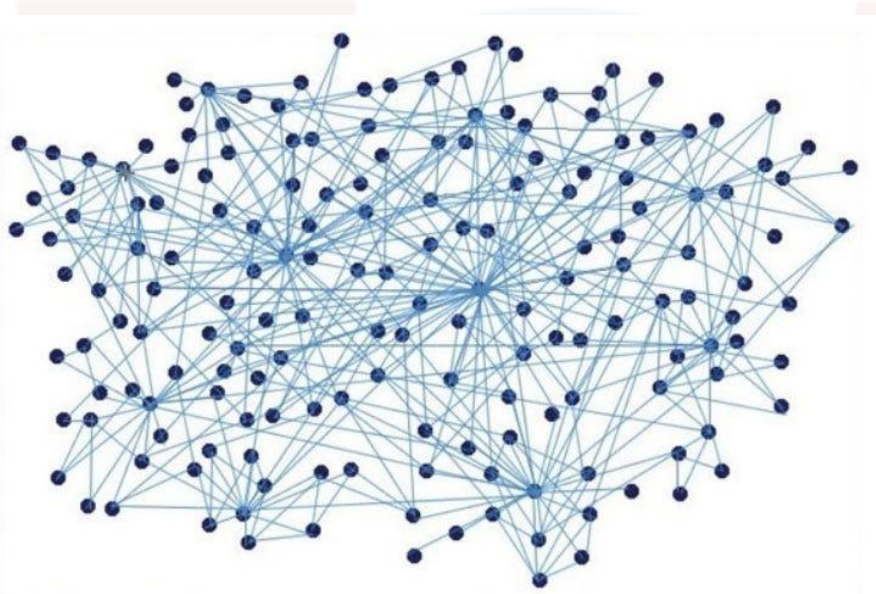
What is a graph?

- A graph is a collection of objects *(nodes)* and their interactions *(edges)*
  - e.g., in a Social Network each person is a node, and an edge between two persons indicate that they are friends
- Why the graph formalism is powerful?
  - to leverage relationships between data points
  - it can be applied to many different domains
  - graphs offer math foundations to analyze, understand and learn complex systems



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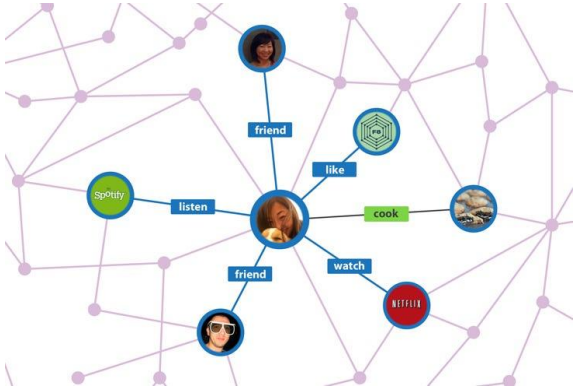
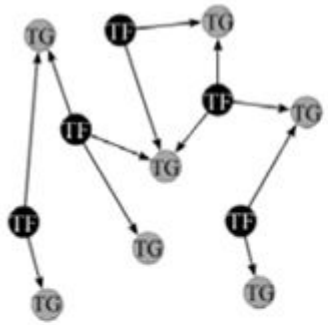
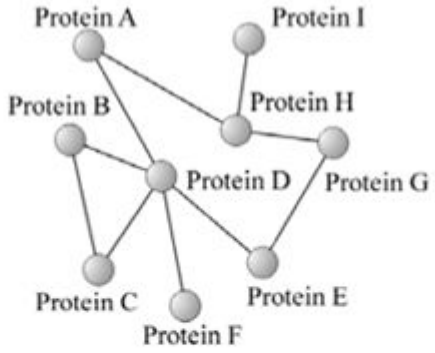
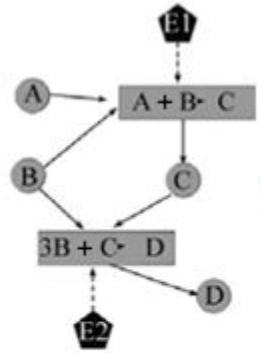
Graphs are everywhere



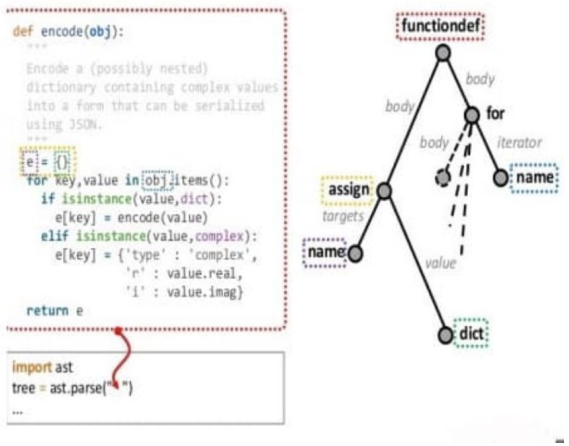
Social networks



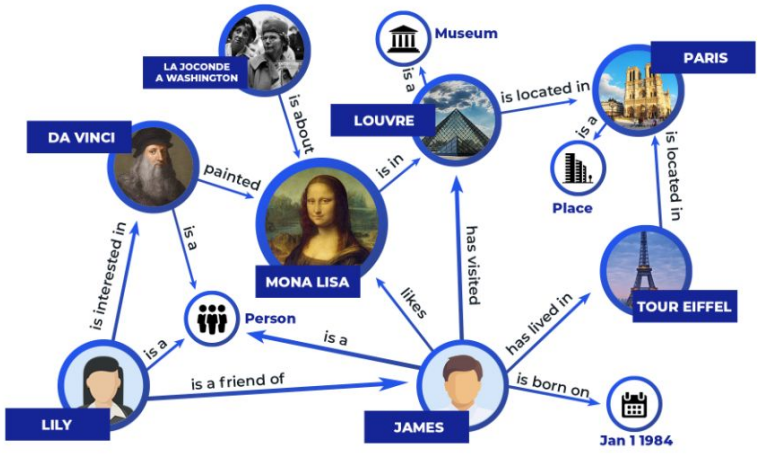




Biology



Software



Knowledge Graphs

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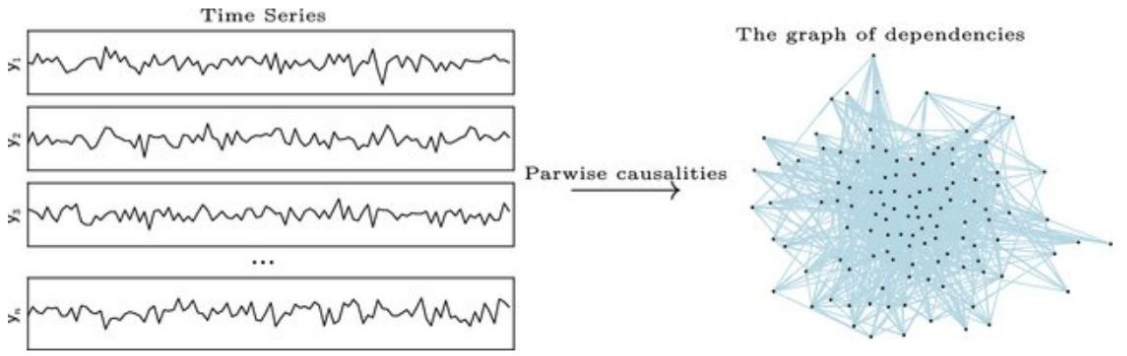
Graphs and Time Series?

- Recently, several research groups proposed ML models to process time series encoded as graphs
- Why?
  - they can capture connections between variables of a multivariate time series
  - they can capture dependencies between different points in time
- This lecture will include the following:
  - basic machine learning tasks with graphs
  - introduction to GNNs
  - how to apply GNNs to time-series
  - applications in Ambient Intelligence



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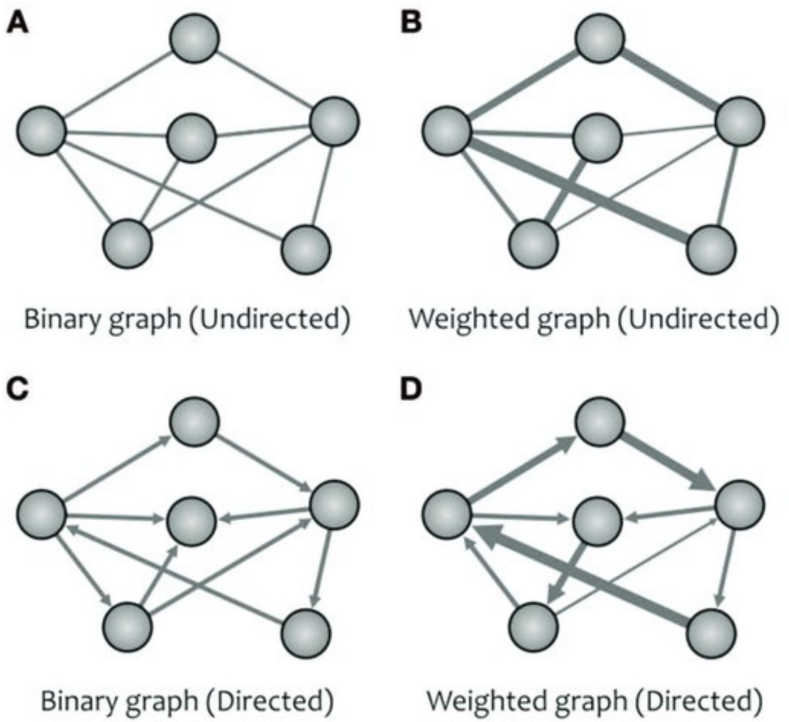
Preliminaries on graphs

Graph definition

- A graph $G = (\boldsymbol{V,E})$ is represented by the set of nodes **V** and set of edges **E**
- If there is an edge from a node $u \in \mathbf{V}$ to another node $v \in \mathbf{V}$, then $(u,v) \in \mathbf{E}$
- Types of graphs:
  - ***undirected***
    - if $(u,v) \in \mathbf{E}$ then $(v,u) \in \mathbf{E}$
  - ***directed***
    - an edge $(u,v) \in \mathbf{E}$ goes from node u to node v
  - ***weighted***
    - each edge is associated with a weight
    - both undirected and directed graphs may be weighted



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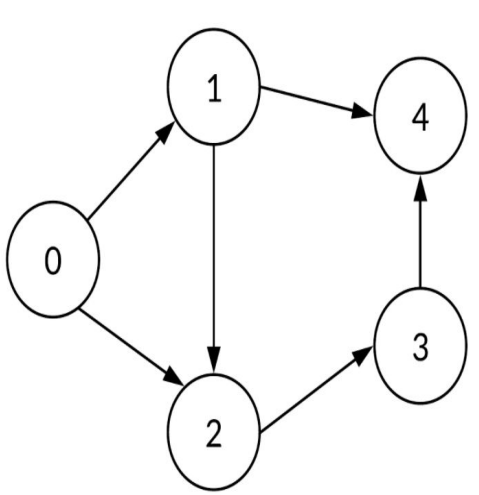
Adjacency Matrix

- Graphs are represented by the Adjacency Matrix $A \in \mathfrak{R}^{|V|*|V|}$
  - A[u,v] = 1 if (u,v)$\in$**E**
  - 0 otherwise
- There is an explicit ordering so that every node identifies a particular row and column
- If the graph is undirected, the matrix is symmetric
- If the graph is directed (i.e., the edge direction matters), the matrix may be not symmetric



Adjacency Matrix

	0	1	2	3	4
0	0	1	1	0	0
1	0	0	1	0	1
2	0	0	0	1	0
3	0	0	0	0	1
4	0	0	0	0	0

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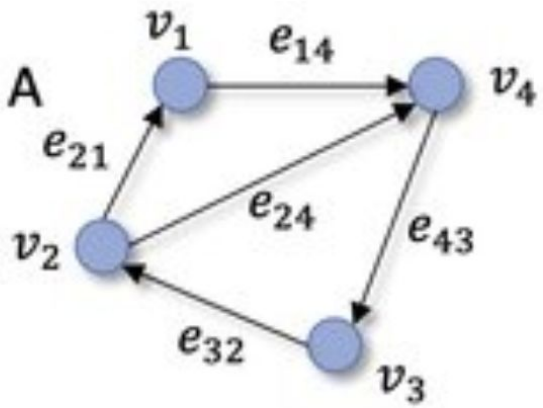
Examples of Adjacency Matrices

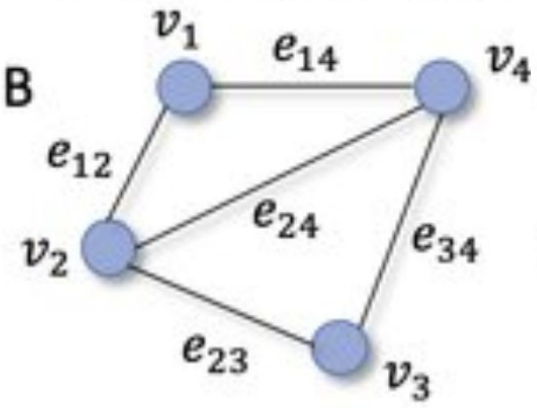
Directed graph G(V,E)



E

	v_1	v_2	v_3	v_4
v_1	0	0	0	1
v_2	1	0	0	1
v_3	0	1	0	0
v_4	0	0	1	0

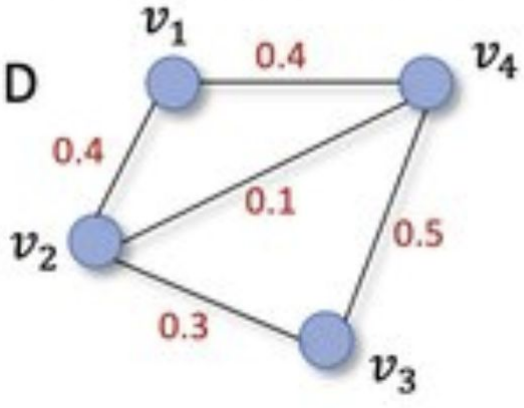
Undirected graph G(V,E)



F

	v_1	v_2	v_3	v_4
v_1	0	1	0	1
v_2	1	0	1	1
v_3	0	1	0	1
v_4	1	1	1	0

Weighted graph G(V,E)



H

	v_1	v_2	v_3	v_4
v_1	0	0.4	0	0.4
v_2	0.4	0	0.3	0.1
v_3	0	0.3	0	0.5
v_4	0.4	0.1	0.5	0

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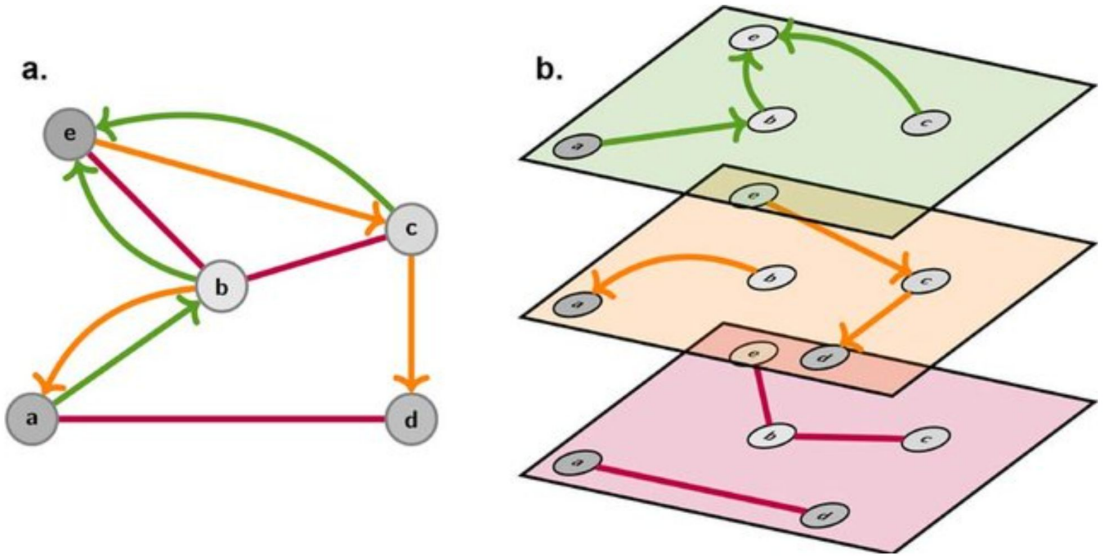
Multi-relational graphs

- Extension of graphs where each relation also has a type
- $(u,\tau,v) \in \mathbf{E} \rightarrow$ nodes *u* and *v* are connected by a relation of type τ
- There is one adjacency A_τ matrix for each relation τ
- Overall adjacency matrix A
  - $A \in \mathfrak{R}^{|V|*|R|*|V|}$
    - where R is the set of all the possible relations



Heterogeneous Graphs

- Multi-relational graphs where also the nodes are typed
- The set of nodes **V** is partitioned in subsets
- $\mathbf{V} = V_1 \cup V_2 \cup \ldots \cup V_k$
  - $\forall\ i \neq j\ V_i \cap V_j = \varnothing$
- Each partition V_i only includes nodes of the same type
- Defining edges also implies having constraints on node types
  - e.g., $(u,\boldsymbol{\tau},v) \in \mathbf{E}$ implies that $u \in V_j$ and $v \in V_k$
- Example:
  - node types: proteins, drugs, and diseases
  - the relationship “treatment” is defined only between drugs and disease nodes

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Node-level Attributes

- In a graph, each node is usually associated with a set of features
  - These are referred as ***node-level attributes***
- The overall node-level attributes of a graph can be represented as
  - $\mathbf{X} \in \mathfrak{R}^{|V|*m}$
    - where *m* is the dimensionality of the node-level attributes
    - assumption: the ordering of nodes is consistent with the ones in the adjacency matrix
- In heterogeneous graphs, each node type has its own distinct attributes!

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Machine Learning on Graphs

Machine Learning tasks for Graphs

- The most popular machine learning tasks on graphs are:
  - Node Classification
  - Link Prediction
  - Graph classification/regression/clustering
- In the following, these tasks will be described

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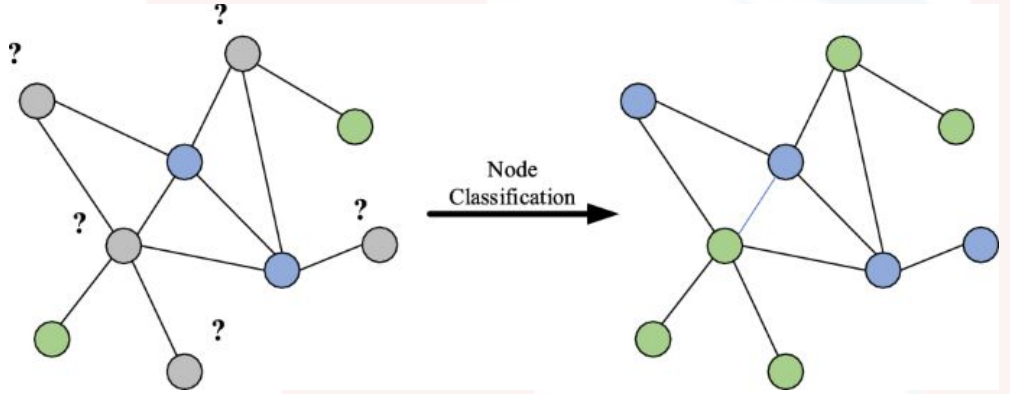
Node Classification

- **Goal:** predict the label y_u for each $u \in \mathbf{V}$ given that labels are available for $V_{train} \subset \mathbf{V}$
- Two common scenarios:
  - ***semi-supervised learning:*** there are a few labeled nodes in the graph and the goal is to classify the unlabeled ones
  - ***supervised learning:*** there are many labeled nodes and the goal is to generalize across disconnected graphs
- Challenge: nodes may be non-IID and connections between them should be leveraged
- Examples of applications:
  - identify bots in social networks, classify documents topic based on hyperlinks



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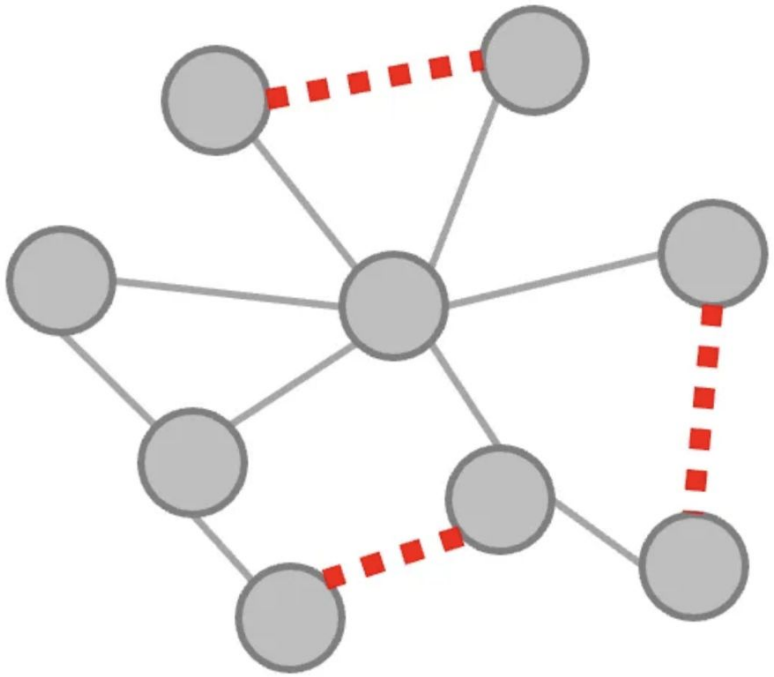
Link Prediction

- Given a set of nodes **V** and an *incomplete* set of edges $E_{train} \subset \mathbf{E}$, the goal is to infer the missing edges
- Example of application:
  - suppose that we have a graph of interactions between proteins, but we only know only some of them. The goal is to infer unknown interactions



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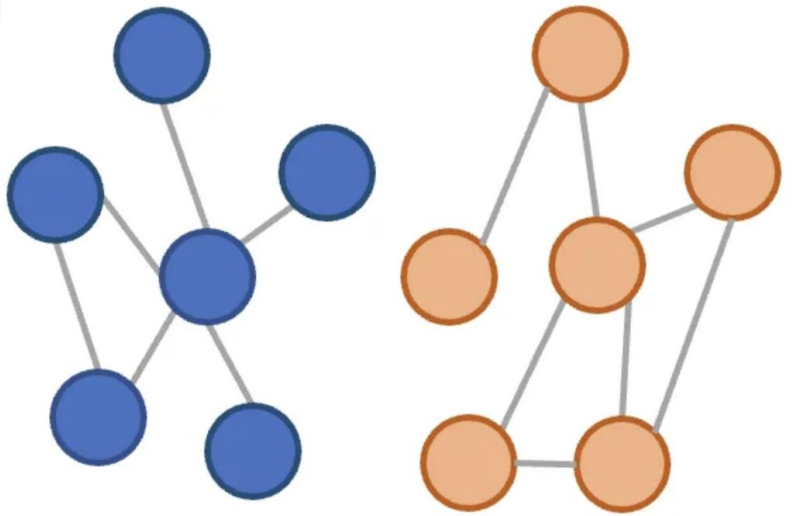
Graph classification/regression

- Until now, the tasks involved predictions on individual components (i.e., edges, nodes)
- Given a dataset of multiple different graphs, it is also possible to make independent predictions on each graph (classification or regression)
- Similarly to standard supervised learning, each graph is a data point associated with a label, and the goal is mapping data points to labels
- Example of application:
  - given a graph-based representation of syntax and data flow, determine if a computer program is malicious



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Graph Clustering

- The unsupervised counterpart of Graph Classification
- Goal: learning an unsupervised measure of similarity between graphs
- Again, it is the “straightforward” extension of clustering on graph data

Challenges

- While these last two tasks seem to be very similar to standard supervised or unsupervised solutions…
- …the challenge is to find features taking into account the graph relational structure!
- A naive approach:
  - just use the adjacency matrix as input (e.g., flattened and provided to an MLP model)
  - **why doesn't it work?**
    - the adjacency matrix has an arbitrary ordering of nodes
    - this representation is not permutation-invariant

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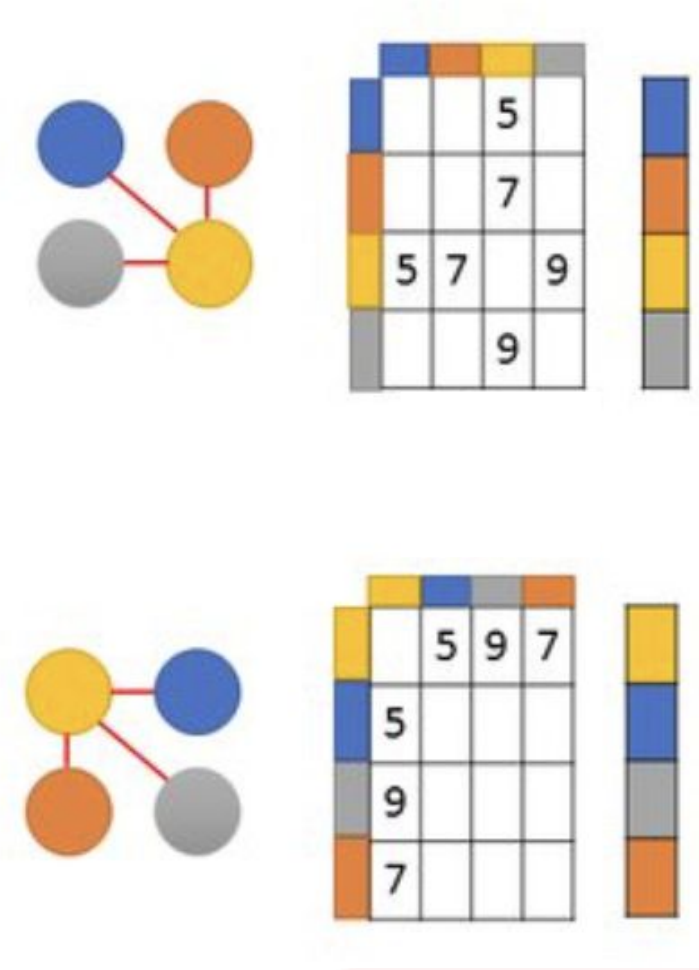
Graph Isomorphism

- These two graphs represent exactly the same thing, even though they reflect two different ***permutations*** of their nodes
  - i.e., they are *isomorphic* to each other
- However, their adjacency matrices are completely different
- The goal of GNNs is to determine a permutation independent graph representation!
  - ***idea:*** the nodes should be considered considered as a *set* where the order is not relevant



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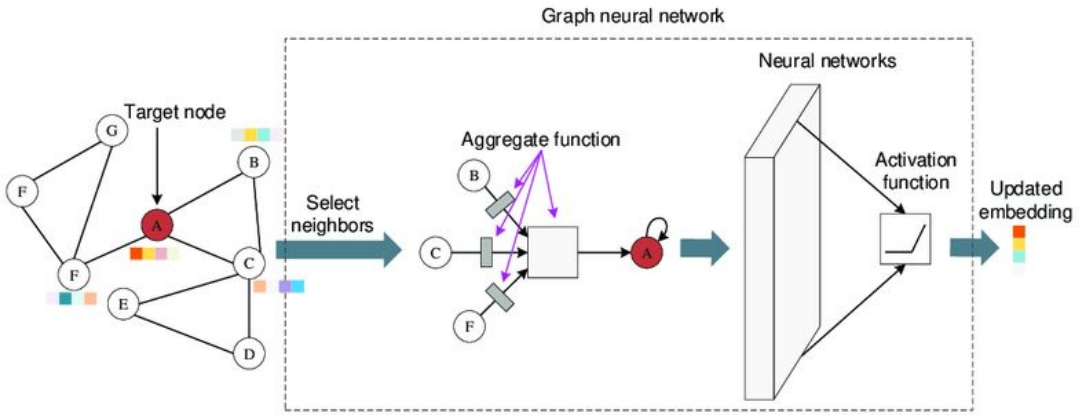
Graph Neural Networks

Graph Neural Networks (GNNs)

- GNNs represent a framework for defining deep neural networks on graph data
- ***Idea***: for each node, learn an embedding representation
- This representation depends:
  - on the structure of the graph
  - on any feature information we might have
- The obtained nodes embedding representation are used to solve the tasks mentioned before



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Message Passing

- The underlying mechanism of GNNs is called *Message Passing*
- It allows updating the embedding of a node by aggregating the representations of its neighbors
- It is based on two functions:
  - ***AGGREGATE:*** it determines how to combine the information from the neighbors
  - ***UPDATE:*** it determines how to update the representation in the target node

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Message Passing: The Formula

Given a node *u,* let $\mathbf{h}_u^{(k)}$ be its representation at the k-th iteration of Message Passing. Let N(*u*) be the neighbors of *u*.

UPDATE and AGGREGATE are arbitrarily differentiable functions (e.g., MLPs).

$\mathbf{m}_{\mathcal{N}(u)}^{(k)}$ is the message aggregated from the neighbors

Aggregation works on a *set* to deal with permutation invariance.

$$\mathbf{h}_u^{(k+1)} = \text{UPDATE}^{(k)}\left(\mathbf{h}_u^{(k)}, \text{AGGREGATE}^{(k)}(\{\mathbf{h}_v^{(k)}, \forall v \in \mathcal{N}(u)\})\right)$$

$$= \text{UPDATE}^{(k)}\left(\mathbf{h}_u^{(k)}, \mathbf{m}_{\mathcal{N}(u)}^{(k)}\right),$$

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How does it work? An intuition

- At time 0, each node is represented with its input features
- $\mathbf{h}^{(0)}_u = x_u \quad \forall u \in \mathbf{V}$
- At each iteration, each node aggregates information from direct neighbors
- As iterations progress, each node contains more and more infos from distant nodes
  - *e.g.,* at k=1 infos from 1-hop neighbors, at k=2 infos from 2-hops neighbors, etc
- At the end (e.g., after K iterations), each node is associated with an embedding encoding features of the K distant neighbors!
  - this is somehow analogous to CNN layers. Instead of spatial-defined patches, the GNN aggregates nodes from local neighbors!

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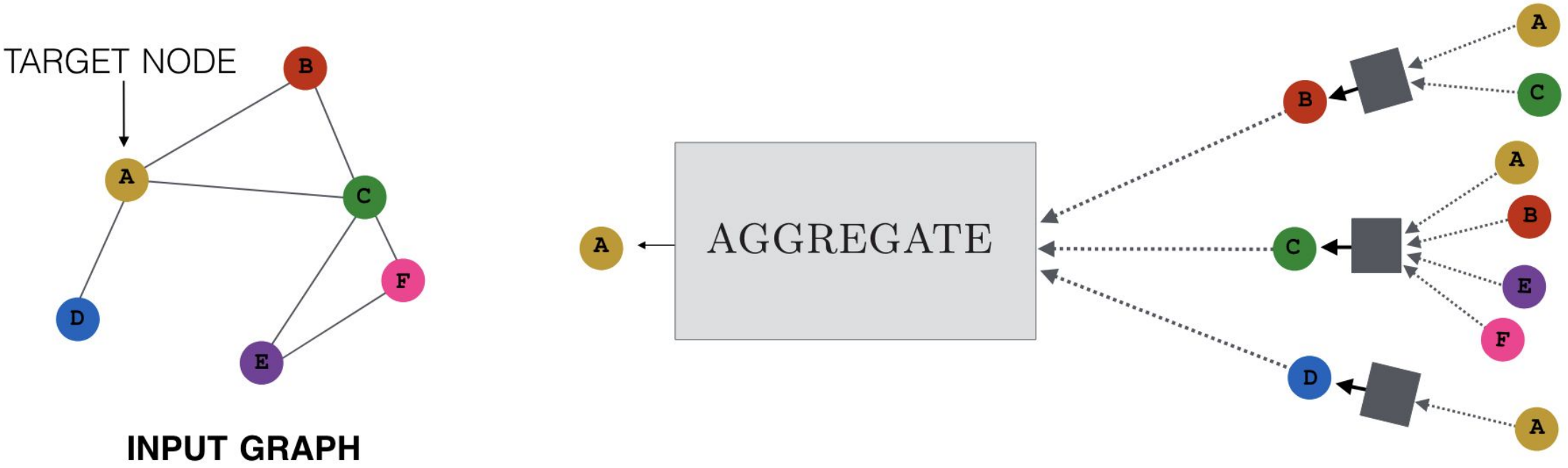
Message Passing visualized



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Naive Message Passing

Two learnable weights for each node.

Using a non-linear activation function (e.g., ReLU) on a linear combination.

Aggregation is simply the sum of neighbors representations.

$$\mathbf{h}_u^{(k)} = \sigma\left(\mathbf{W}_{\text{self}}^{(k)}\mathbf{h}_u^{(k-1)} + \mathbf{W}_{\text{neigh}}^{(k)} \sum_{v \in \mathcal{N}(u)} \mathbf{h}_v^{(k-1)} + \mathbf{b}^{(k)}\right)$$

In the following slides, the bias term will be omitted for simplicity.

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Naive Message Passing: AGGREGATE and UPDATE

$$\text{AGGREGATE}^{(k)}(\{\mathbf{h}_v^{(k)}, \forall v \in \mathcal{N}(u)\}) = \mathbf{m}_{\mathcal{N}(u)} = \sum_{v \in \mathcal{N}(u)} \mathbf{h}_v$$

$$\text{UPDATE}(\mathbf{h}_u, \mathbf{m}_{\mathcal{N}(u)}) = \sigma\left(\mathbf{W}_{\text{self}}\mathbf{h}_u + \mathbf{W}_{\text{neigh}}\mathbf{m}_{\mathcal{N}(u)}\right)$$

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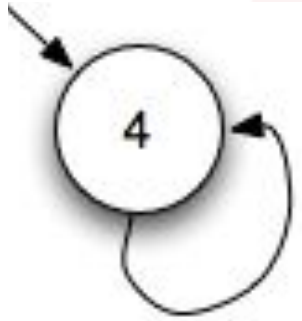
Message Passing with Self-Loops

Adding self-loops to each node for simplification, so to remove the UPDATE step!

$$\mathbf{h}_u^{(k)} = \text{AGGREGATE}(\{\mathbf{h}_v^{(k-1)}, \forall v \in \mathcal{N}(u) \cup \{u\}\})$$

Pros: it simplifies message passing, often alleviating overfitting. Only the weight for aggregation should be learned by the GNN.

Cons: it limits the expressivity of GNNs, since it's not possible to differentiate infos from neighbors with infos from the node itself!



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Normalization

- In the previous examples, aggregation was quite simple and based on the sum neighbors embeddings
- However, this may be unstable!
  - e.g. if a node *u* has 100x the number of neighbors than a node *u',* the sum may lead to drastic differences in magnitude!
- It is possible to perform normalization, or *symmetric* normalization to mitigate this problem!

$$\mathbf{m}_{\mathcal{N}(u)} = \frac{\sum_{v \in \mathcal{N}(u)} \mathbf{h}_v}{|\mathcal{N}(u)|}$$

Normalization

$$\mathbf{m}_{\mathcal{N}(u)} = \sum_{v \in \mathcal{N}(u)} \frac{\mathbf{h}_v}{\sqrt{|\mathcal{N}(u)||\mathcal{N}(v)|}}$$

Symmetric Normalization

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Graph Convolutional Networks (GCNs)

Graph Convolutional Networks (GCNs)

Basically, GCNs combine self-loops and symmetric normalization.

$$\mathbf{h}_u^{(k)} = \sigma\left(\mathbf{W}^{(k)} \sum_{v \in \mathcal{N}(u) \cup \{u\}} \frac{\mathbf{h}_v}{\sqrt{|\mathcal{N}(u)||\mathcal{N}(v)|}}\right)$$

Convolution!

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Set Pooling

- Instead of summing, it is possible to do something more advanced for the **aggregation**
  - i.e., mapping a *set* of embeddings into a single embedding
- Known results in ML show that a permutation-invariant function mapping a set of embeddings to a single embedding can be approximated using the following formula (called *universal set function approximator*):

$$\mathbf{m}_{\mathcal{N}(u)} = \mathrm{MLP}_\theta \left(\sum_{v \in N(u)} \mathrm{MLP}_\phi(\mathbf{h}_v) \right)$$

- Here, MLP is an arbitrary deep multi-layer perceptron in charge of generating a latent representation of the input

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Janossy Pooling

- Alternative to Set Pooling, proved to be more powerful
- It leverages permutation-sensitive functions
- Goal: averaging over ***many*** possible permutations
- Let $\pi_i \in \mathbf{\Pi}$ be a permutation taking the unordered set of neighbor embeddings and placing these embeddings in a specific sequence based on some arbitrary ordering.

$$\mathbf{m}_{\mathcal{N}(u)} = \mathrm{MLP}_\theta \left(\frac{1}{|\Pi|} \sum_{\pi \in \Pi} \rho_\phi \left(\mathbf{h}_{v_1}, \mathbf{h}_{v_2}, \ldots, \mathbf{h}_{v_{|\mathcal{N}(u)|}} \right)_{\pi_i} \right)$$

ρ_ϕ is a permutation-sensitive function extracting embeddings (e.g., LSTM).

Since trying all the possible permutations is intractable, these are usually sampled.

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Graph-Level Embedding: *Graph Pooling*

- Message passing leads to a set of node embeddings
- What if we want to obtain a global latent representation of the graph?
- ***Goal:*** given nodes embeddings, $\forall z_u \in \mathbf{V}$, how to learn a representation z_G for the entire graph G?
- *Learning over sets:*

$$\mathbf{z}_{\mathcal{G}} = \frac{\sum_{v \in \mathcal{V}} \mathbf{z}_u}{f_n(|\mathcal{V}|)}$$

normalization function

Note that this may be further improved by using an attention mechanism, in charge of learning a weight for each z_u and performing a weighted sum!

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Let’s go back to our tasks!

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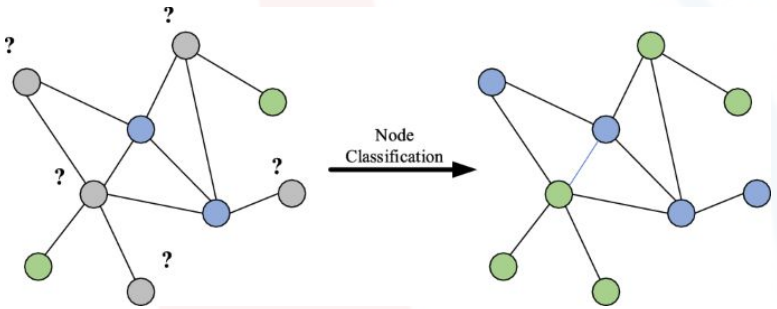
Node Classification

- Performing classification on each node based on the embeddings obtained after message passing
- In *transductive* scenarios, message passing can also leverage unlabeled nodes (semi-supervised)
- It is however crucial to ignore test nodes (also called inductive nodes) during training!
  - these nodes should not impact neither message passing and the training phase
- A possible loss function:

$$\mathcal{L} = \sum_{u \in \mathcal{V}_{\text{train}}} -\log(\text{softmax}(\mathbf{z}_u, \mathbf{y}_u)).$$



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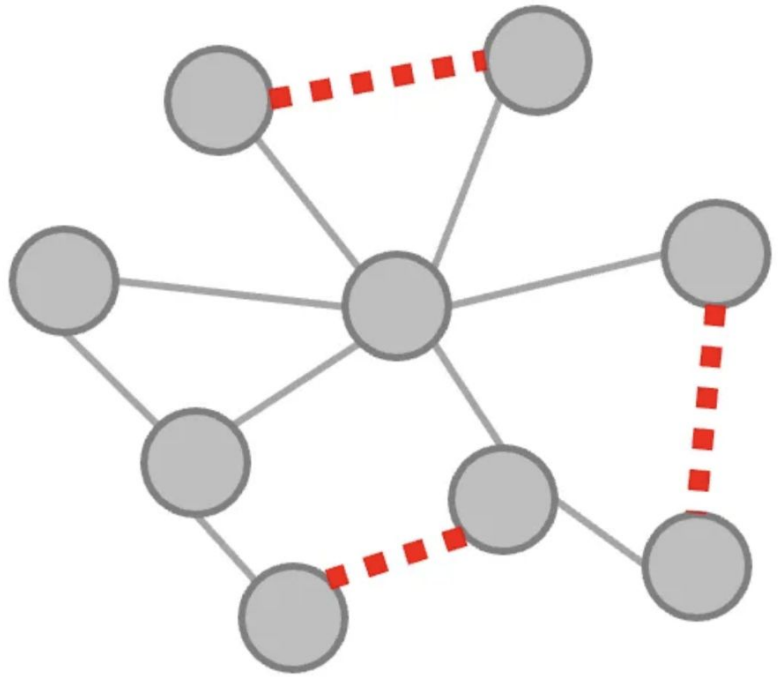
Link Prediction

- Link prediction is usually framed as a metric learning problem:
  - minimizing the distance between the embeddings of nodes collected by a link
  - maximizing the distance between the embeddings of nodes not collected by a link
- Hence, a link is predicted between two nodes if their embeddings are similar (e.g., based on a threshold!)



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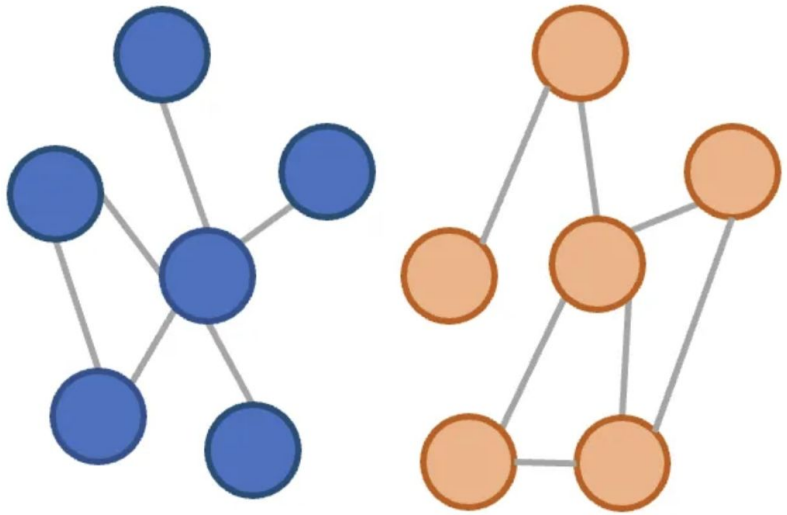
Graph Classification/Regression/Clustering

- This is the “simplest” case!
- It is sufficient to obtain the graph embedding z_G for each graph in the dataset, after graph pooling
- Then, it is possible to apply standard DL approaches
  - i.e., each Z_G is a data point



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GNNs in Ambient Intelligence Scenarios

GNN in Ambient Intelligence

- There are several benefits of applying a GNN to model sensor data, besides what is provided by CNN and RNN
- CNNs and RNNs can somehow be viewed as GNNs
  - i.e., fixed-size grid graphs for CNNs and line graphs for RNNs
- Typical GNNs deal with complex graphs
  - no fixed form
  - variable size of unordered nodes
  - variable amount of neighbors for each node
- Advantages of GNNs:
  - effectively encode complex relationships and interdependencies among devices in IoT sensing systems, and among data instances over time!

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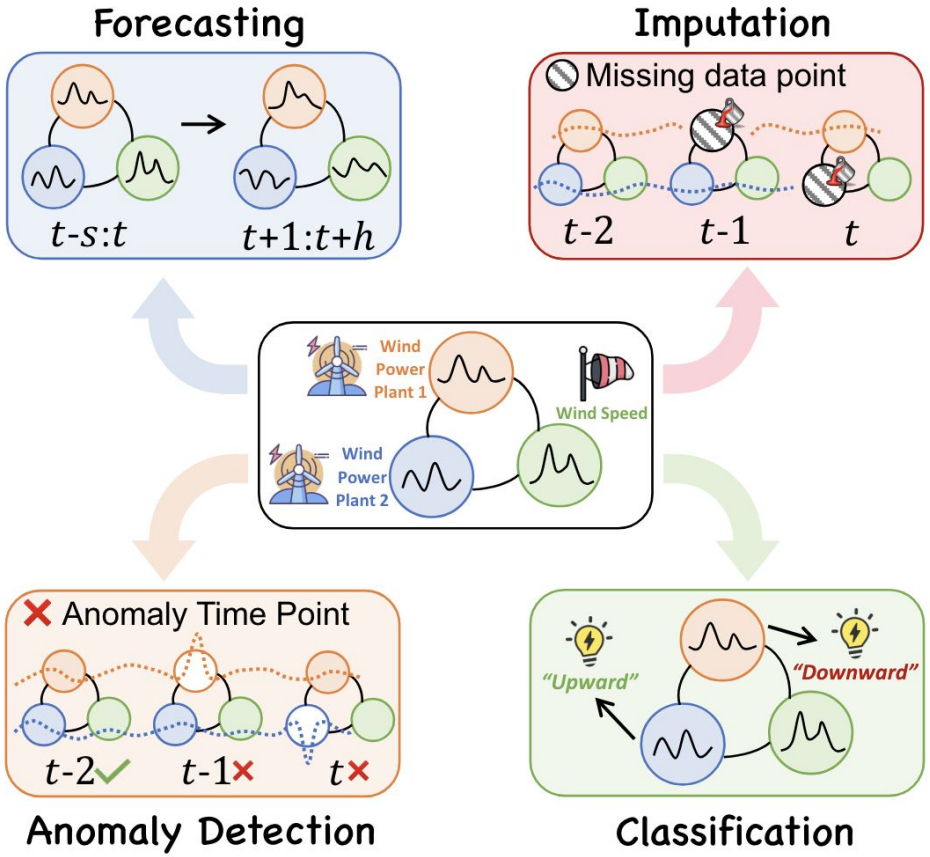
Time series GNN tasks



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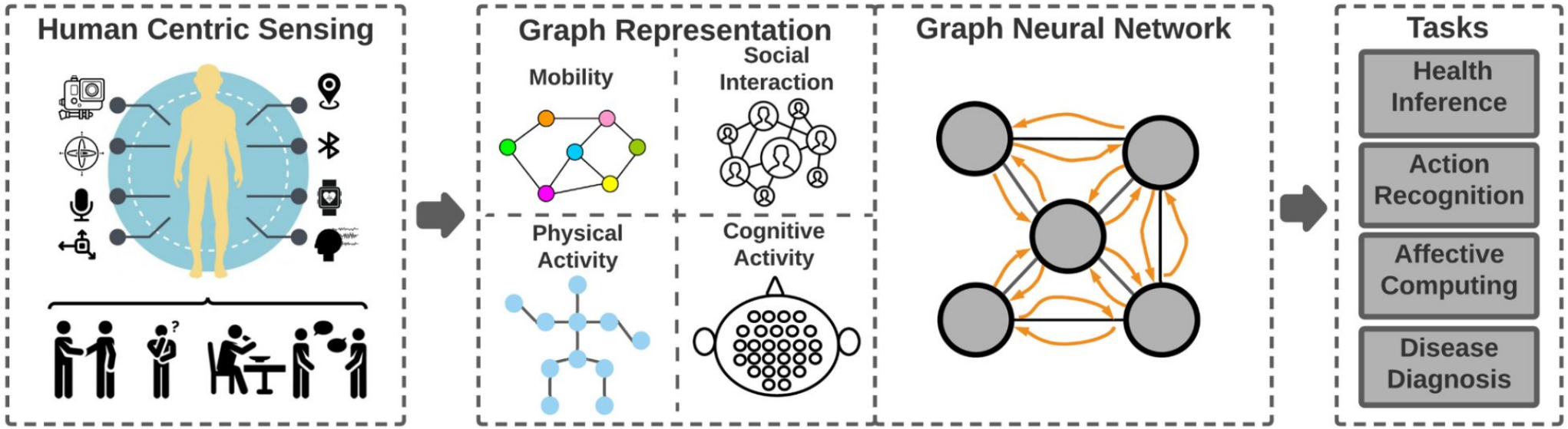
GNNs in Human Centric Sensing



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Graph construction in Time Series

- How to construct a graph from a multivariate time series?
- Each node may represent a sensor (or a device with multiple sensors)
  - node features: a time window of sensor data
- How to set the edges between nodes?
  - **Heuristic-based graphs:**
    - using some heuristics like spatial proximity between sensors, pairwise similarity between sensors based on training data, etcetera
  - **Learning-based graphs:**
    - edges are learned in a end-to-end fashion. Initially, there is an edge between each pair of nodes
    - attention-based mechanisms can be used to derive the most important edges
    - more costly than heuristic-based, but it may lead to complex and informative graph structures

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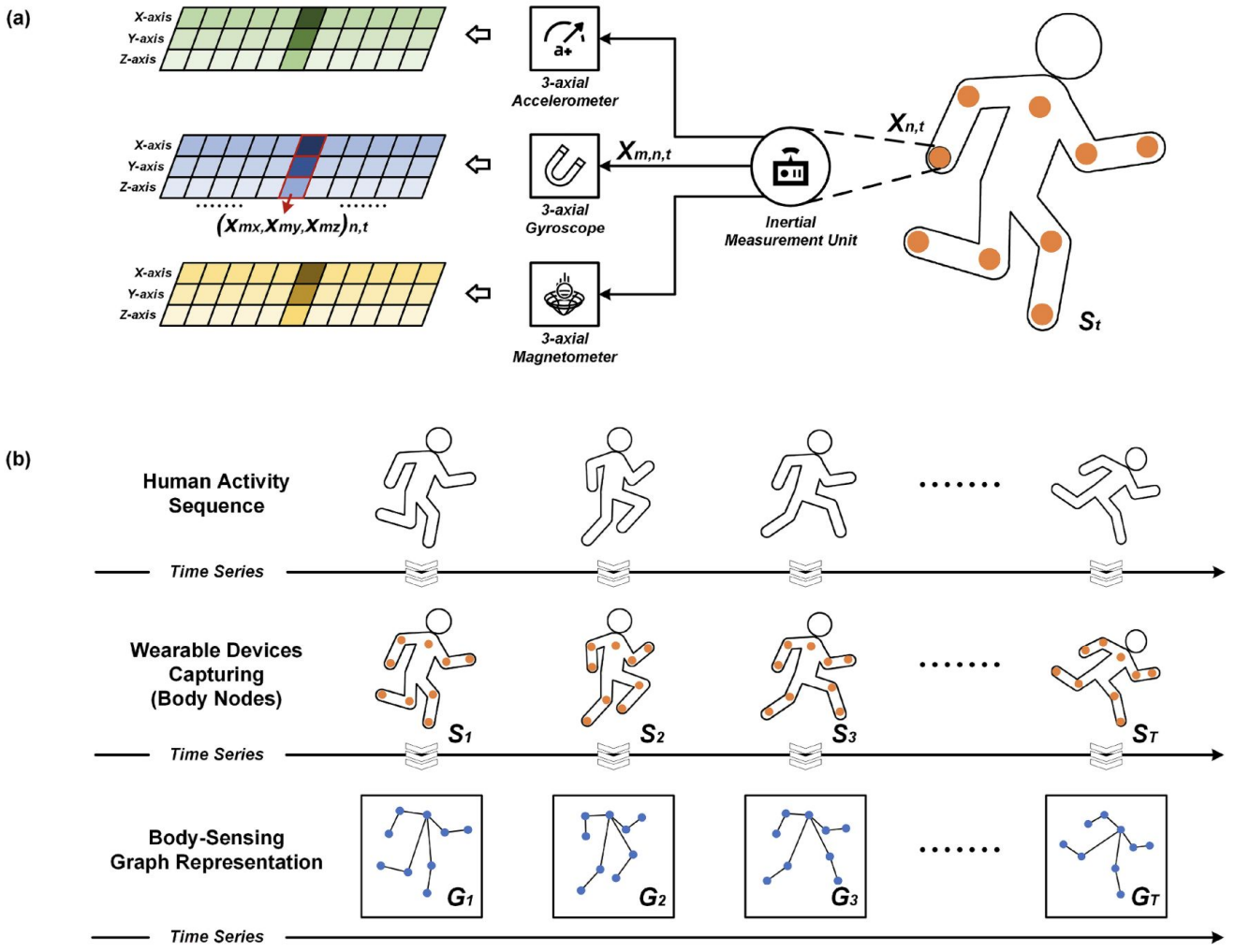
Body-sensing Graph Representation for wearable-based HAR



Zou, H., Chen, Z., Zhang, J., Wang, L., Zhang, F., Li, J., & Pan, Y. (2024). GT-WHAR: A Generic Graph-Based Temporal Framework for Wearable Human Activity Recognition With Multiple Sensors. *IEEE Transactions on Emerging Topics in Computational Intelligence*.

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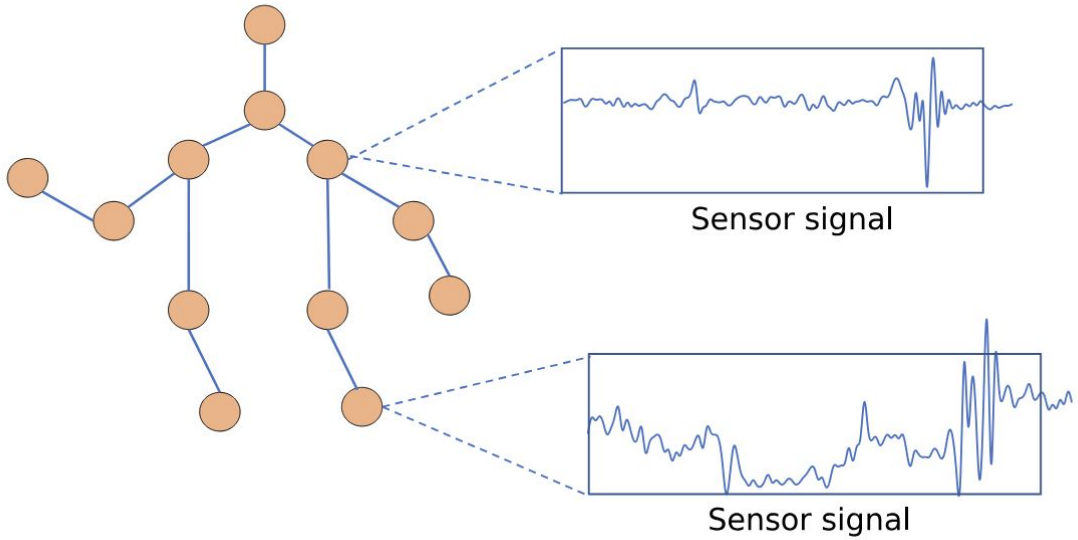
Leveraging spatial relationships

Each node represents a device (possibly with multiple sensors) positioned in a body position.

Feature node: a time window of sensor data collected in that position.

Edges directly connect adjacent body parts to capture dependencies between close devices!



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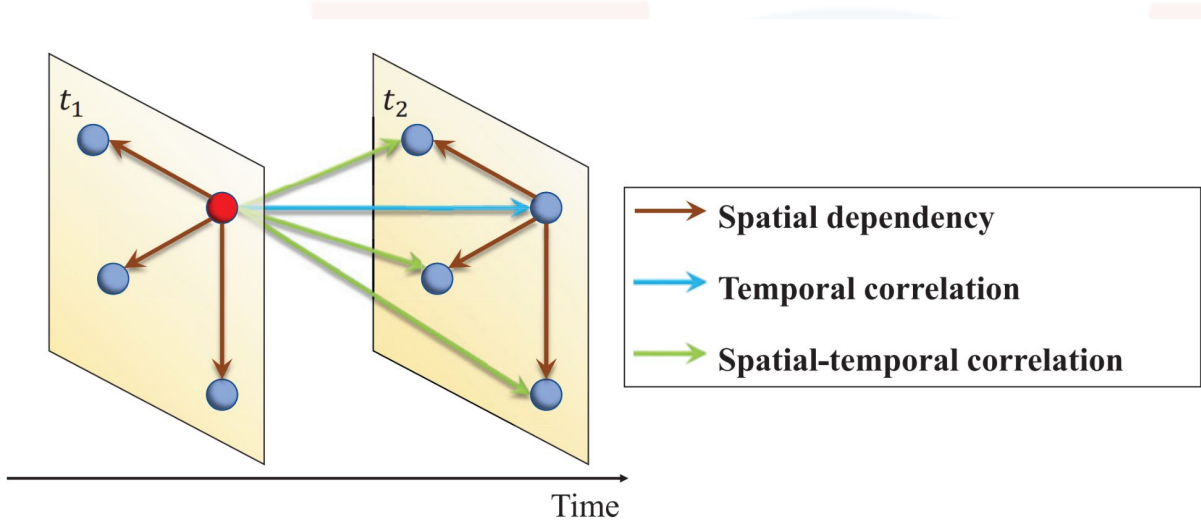
SpatioTemporal Graphs

- Why not encoding both space and time in the graph?
- ***Idea:*** Multi-relational graph, where different layers represent different time instants
  - the challenge is to define what an “instant” is (e.g., one second of sensor data data)
- ***Spatial dependency:*** Nodes connected by an edge at the same time instant
- ***Temporal correlation:*** The same node connected at different time instants
- **Spatial-temporal correlation:** Different nodes connected in different time instants



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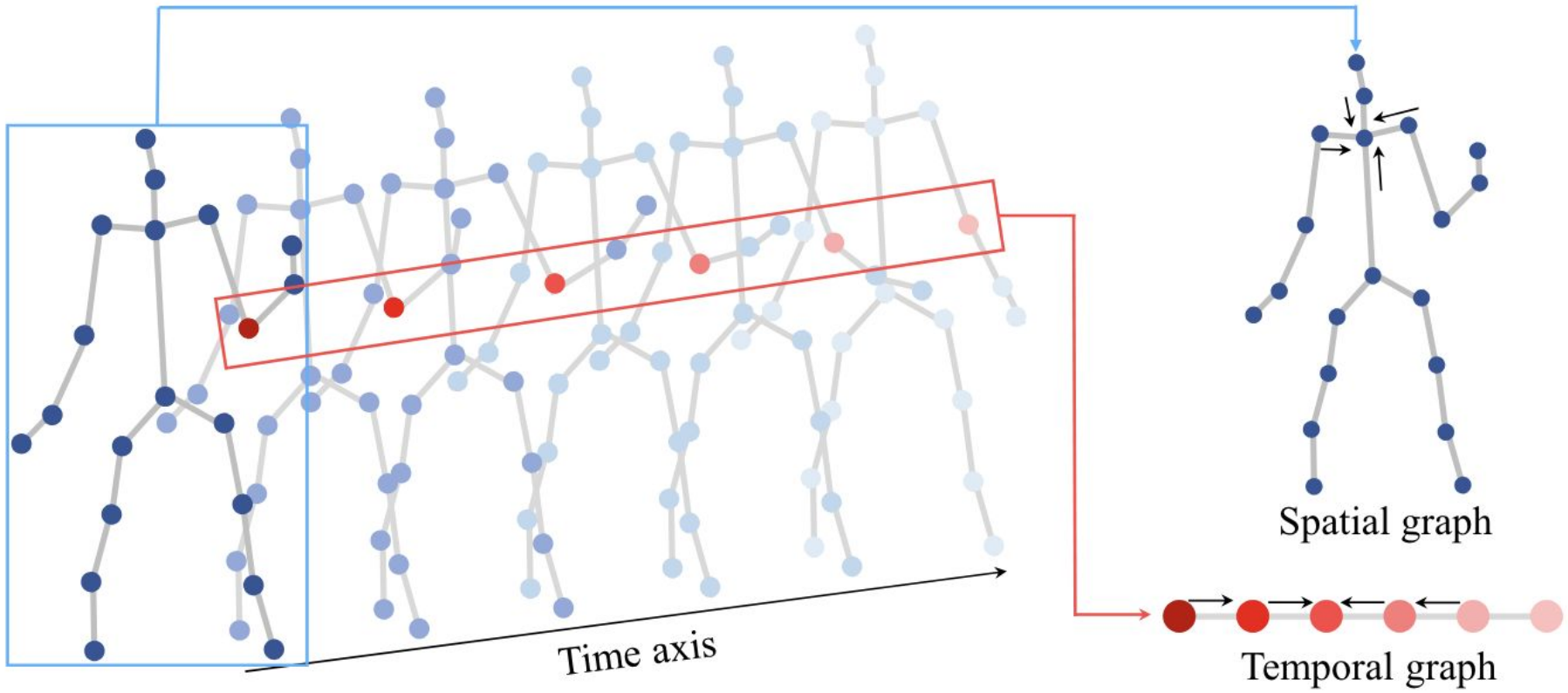
SpatioTemporal Graph with body-sensing graph representation



Li, M., Chen, S., Zhao, Y., Zhang, Y., Wang, Y., & Tian, Q. (2021). Multiscale spatio-temporal graph neural networks for 3d skeleton-based motion prediction. *IEEE Transactions on Image Processing*, *30*, 7760-7775.

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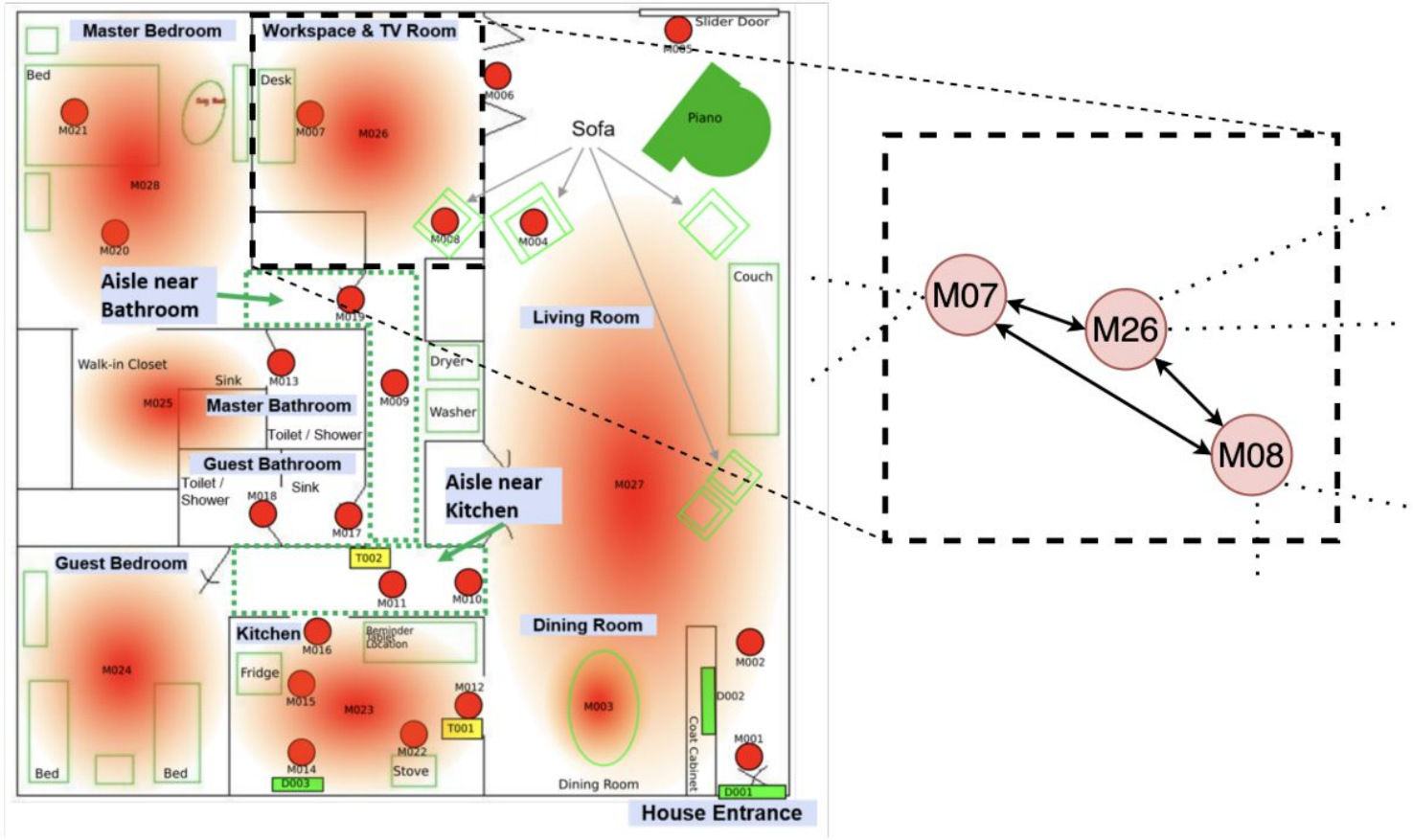
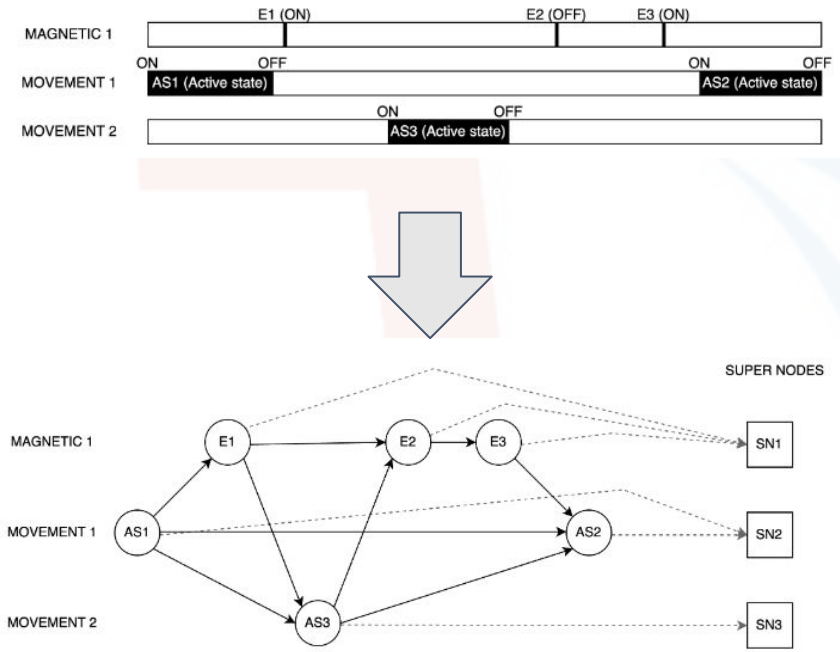
GNNs for Smart-Home HAR



Graph construction in Smart-Homes (heuristic-based)

- In this approach, a window of sensor events is converted into a graph by encoding the temporal relationships between events
- Main idea:
  - each node (i.e., a sensor event) is connected with an edge to another node if they are temporally consecutive in the window



Fiori, M., Mor, D., Civitarese, G., & Bettini, C. (2025). GNN-XAR: A Graph Neural Network for Explainable Activity Recognition in Smart Homes. arXiv preprint arXiv:2502.17999.

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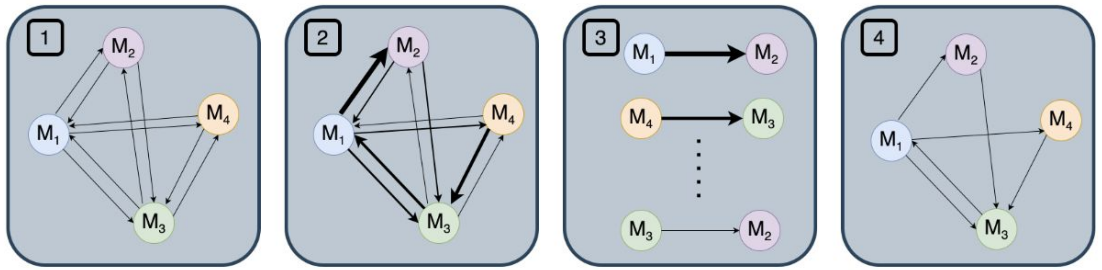
Graph construction in Smart-Homes (heuristic + learning)

- Start with a fully connected graph
  - each node is a sensor
- Compute pairwise similarity based on *sensor behavior*
  - e.g., frequency of activation, periodicity of triggering, range of measured values
- Keep only the edges with the highest similarity!
- Edges are then weighted with an attention mechanism
  - giving higher weights to couple of sensors that are most likely triggered together



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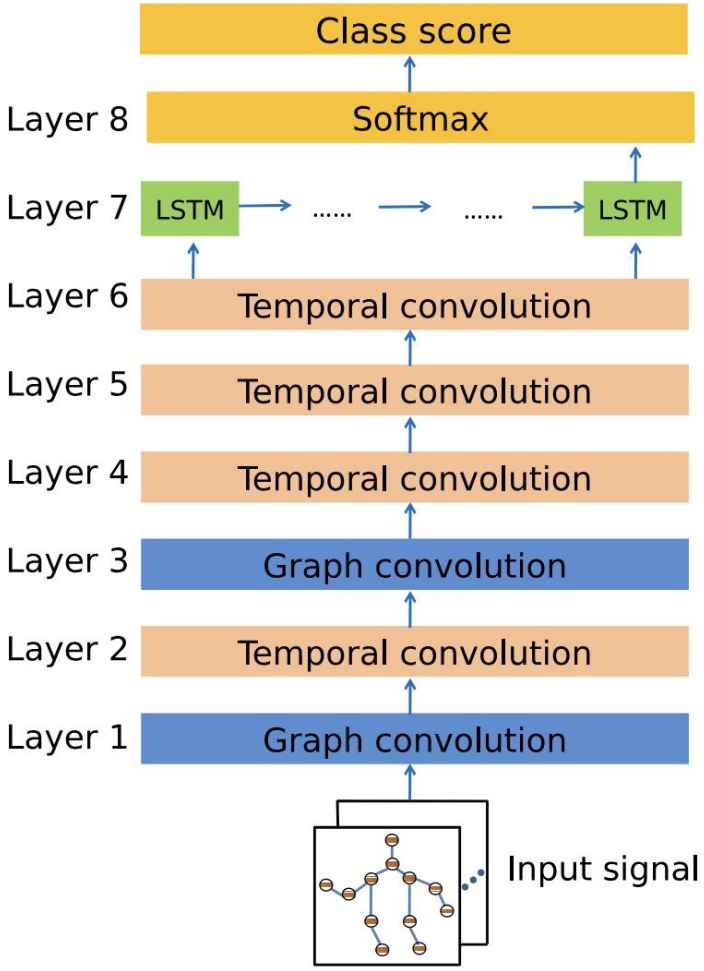
A GNN for Time Series: GraphConvLSTM

- Graph Convolution is computed to capture the *spatial* properties in the graph
- Temporal Convolution is performed using a 1D-CNN layer applied individually on each node, so to capture *temporal* properties
- These two convolutions have the goal of learning node embeddings capturing spatio-temporal properties
- The overall time series is finally processed by LSTM to capture long-term temporal dependencies before classification



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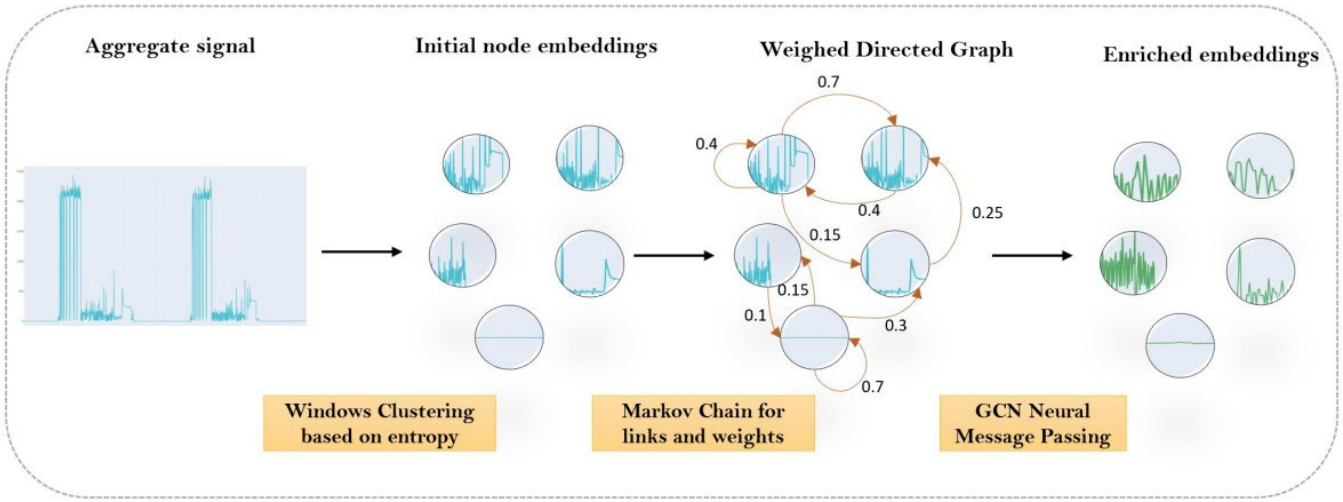
[Han, J., He, Y., Liu, J., Zhang, Q., & Jing, X. (2019, December). GraphConvLSTM: Spatiotemporal learning for activity recognition with wearable sensors. In *2019 IEEE Global Communications Conference (GLOBECOM)* (pp. 1-6). IEEE.]

GNNs for Load Disaggregation

- The aggregated signal from the smart meter is *clustered* to identify different operational modes of appliances
  - each cluster is a node!
- Edges between nodes are weighted, where each weight is the probability of transitioning from one pattern to the other
- GCN is adopted for message passing, and then graph pooling is applied to obtain a single embedding of the aggregated signal
- The graph embedding is provided to a decoder that maps it to power consumption of individual appliances!



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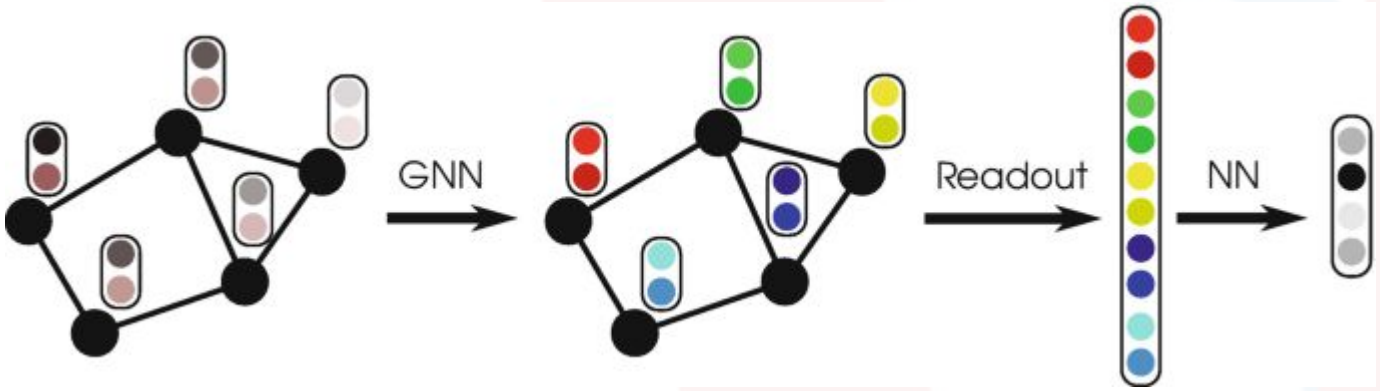
GNNs for Indoor Localization

- Each node represents a specific Access Point
  - the feature value is the measured RSSI
- A weighted edge between nodes is added if the two APs are spatially close
  - by looking at the map…
  - …or automatically, by mining the training set to find instances where the strength signal was high for both APs at the same time
- After message passing, graph pooling is performed to identify the position of the subject from the graph embedding



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Advanced GNNs topics

Advanced GNN topics

- In order to connect this lecture with the previous ones we will answer the following questions:
  - can GNNs be used for self-supervised learning?
  - can we employ XAI methods for GNNs?

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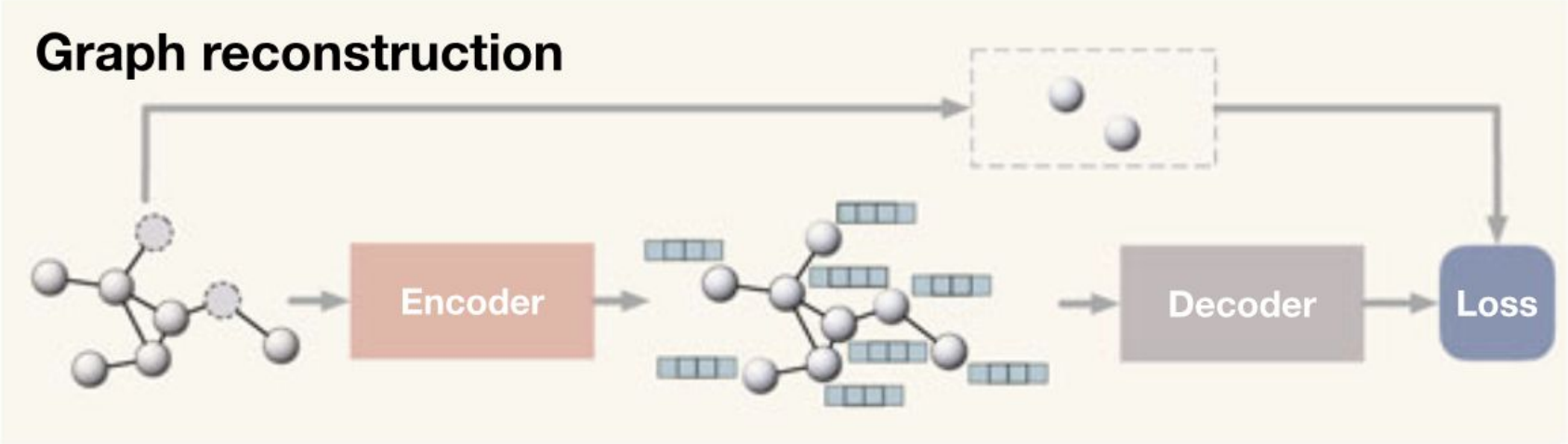
Self-supervised Learning with GNNs: A Predictive Approach



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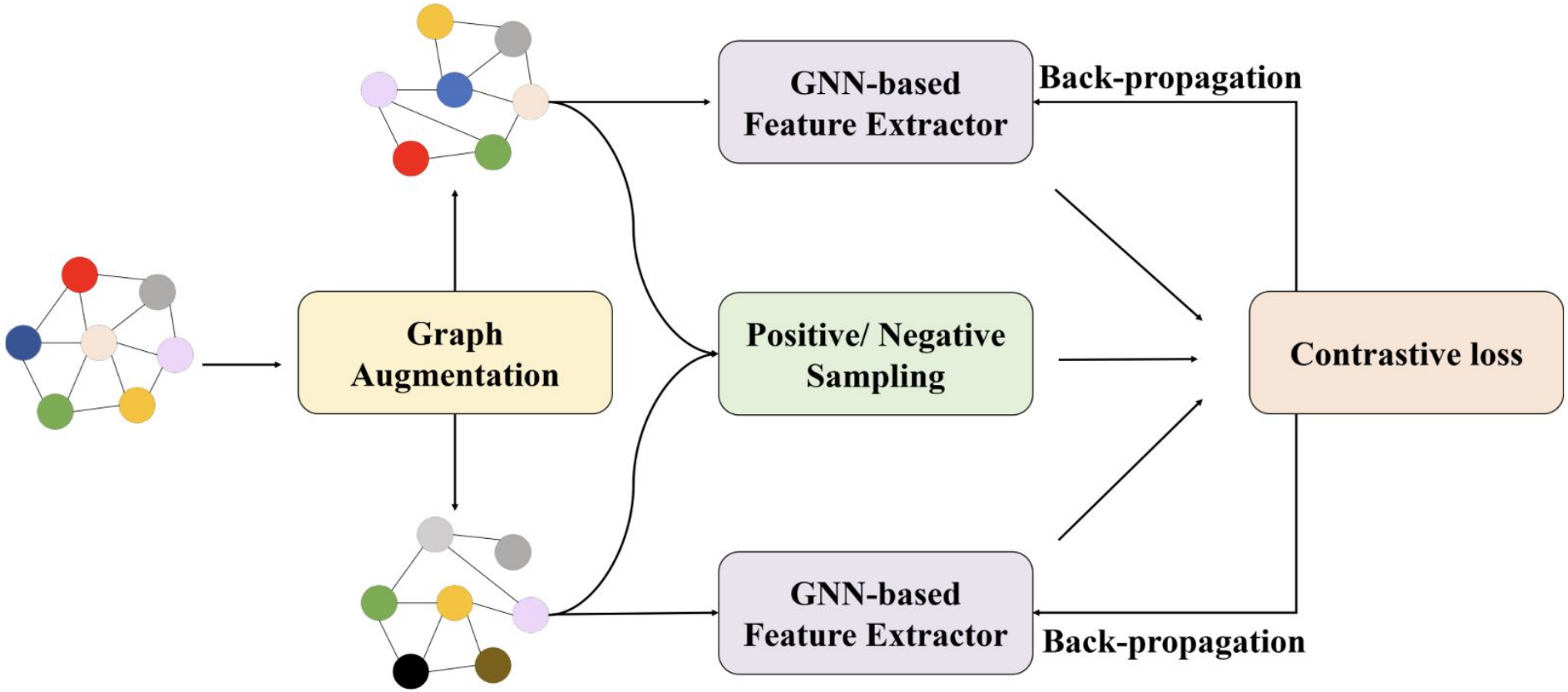
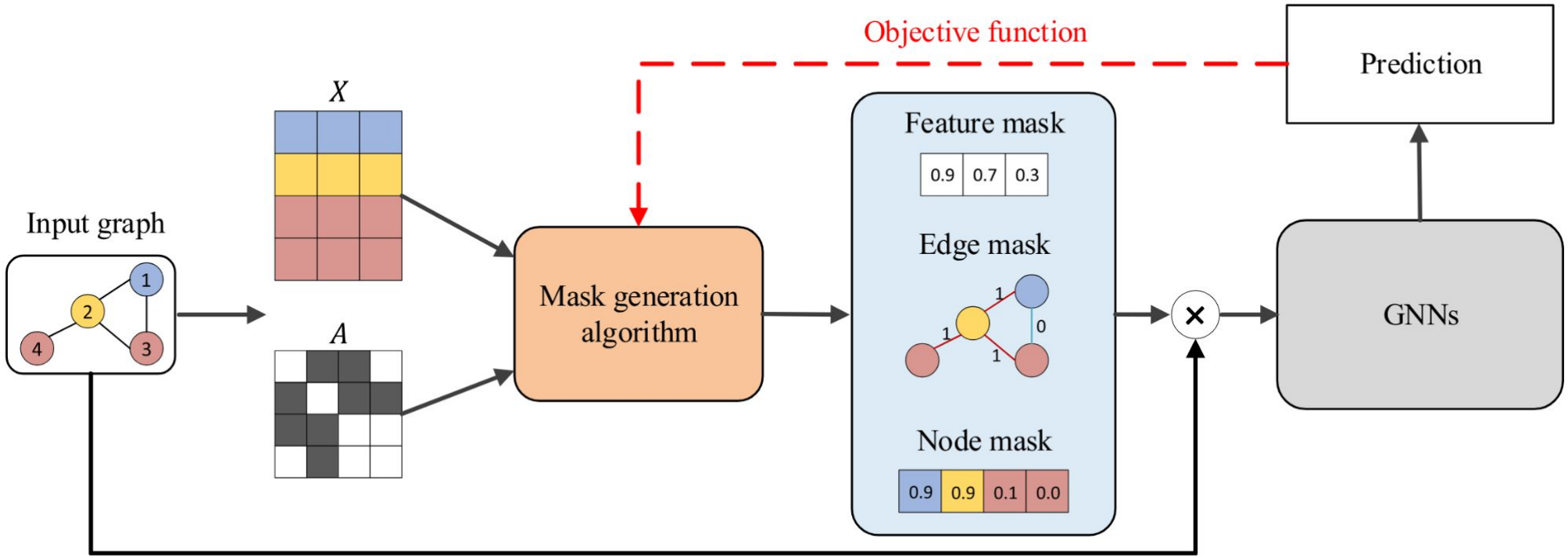
Self-supervised Learning with GNNs: A Contrastive Approach



GNNs and Explainability: A perturbation-based Approach



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